

Skin and Mucous Membrane Adverse Effects

Most patients receiving retinoids develop dryness of the lips, skin, and mucous membranes. These dose-related effects primarily reflect decreased sebum production, reduced stratum corneum thickness, and altered skin barrier function. **Cheilitis** is the earliest and most frequent sign, followed by **blepharoconjunctivitis**, dry eyes, dry nose, and dry mouth. **Xerosis** of the skin, associated with pruritus and peeling—especially on the palms and soles—occurs frequently. Skin fragility and fissuring of the fingertips can be problematic, particularly for individuals who use their hands extensively.

Photosensitivity is commonly observed with **isotretinoin** and is likely due to thinning of the stratum corneum. **Bexarotene** tends to cause fewer mucocutaneous and ocular side effects compared to other retinoids; the most common cutaneous side effect with bexarotene is localized or extensive **exfoliative dermatitis**. *Staphylococcus aureus* colonization often correlates with isotretinoin-induced reduction in sebum production and may progress to overt cutaneous infections. Various nonspecific skin eruptions, collectively termed **retinoid dermatitis**, are frequently seen. The resulting erythema can be difficult to distinguish from underlying **psoriasis** or **atopic dermatitis**.

Ocular Adverse Effects

Blepharoconjunctivitis occurs in approximately one-third of patients treated with isotretinoin, with variable severity, and may interfere with contact lens use. Bacterial conjunctivitis is less common than *S. aureus* colonization. If symptoms persist despite artificial tears or topical antibiotic therapy, ophthalmologic evaluation is recommended. Additional visual disturbances

include **impaired night vision, increased sensitivity to glare, and altered color perception.**

Hair and Nail Adverse Effects

Telogen effluvium, presenting as diffuse or localized hair loss, is a common complaint—more pronounced with **acitretin** than isotretinoin. Objective alopecia typically occurs at higher doses and after several months of treatment. Nail changes such as **thinning** and **paronychia-like lesions** with periungual granulation tissue may also develop.

Musculoskeletal Adverse Effects

Skeletal toxicity has been reported in patients on high-dose, long-term retinoid therapy for keratinization disorders. **Bone pain** without radiographic abnormalities or long-term consequences is relatively common.

Conflicting reports have linked synthetic retinoids to **diffuse idiopathic skeletal hyperostosis (DISH)-like changes**, as well as calcification of tendons and ligaments. However, prospective studies suggest that retinoid-associated hyperostosis is mostly asymptomatic and likely represents progression of pre-existing skeletal overgrowth rather than new-onset disease.

Osteoporosis has been associated with hypervitaminosis A and long-term use of **etretinate**, and to a lesser extent **isotretinoin**. In children, rare cases of skeletal abnormalities and **premature epiphyseal closure** have been documented. Baseline radiographs are not routinely required, but monitoring may be considered in high-risk patients on prolonged, high-dose therapy.

Myalgia and **muscle cramps** are uncommon with etretinate or acitretin but occur more frequently with isotretinoin, especially in individuals engaged in vigorous physical activity—sometimes accompanied by elevated **creatinine**

phosphokinase (CPK) levels. Increased muscle tone, axial muscle rigidity, and myopathy have been reported with etretinate and acitretin use.

Central Nervous System and Other Neurologic Adverse Effects

Central nervous system side effects are rare. While isolated symptoms such as headache, nausea, and vomiting may occur, the full clinical picture of pseudotumor cerebri (increased intracranial pressure with papilledema and visual impairment) is uncommon. The concomitant use of tetracycline antibiotics—which themselves can increase intracranial pressure—is a major risk factor for developing this syndrome in patients taking isotretinoin.

Anecdotal reports have suggested a link between isotretinoin and severe depression or suicidal ideation. However, large-scale epidemiologic studies have not shown an increased risk of psychiatric disorders with isotretinoin compared to other acne treatments, including antibiotics.